

『JISOO』 ENGINE

3팀 김형도 김지수 이준협 성원희



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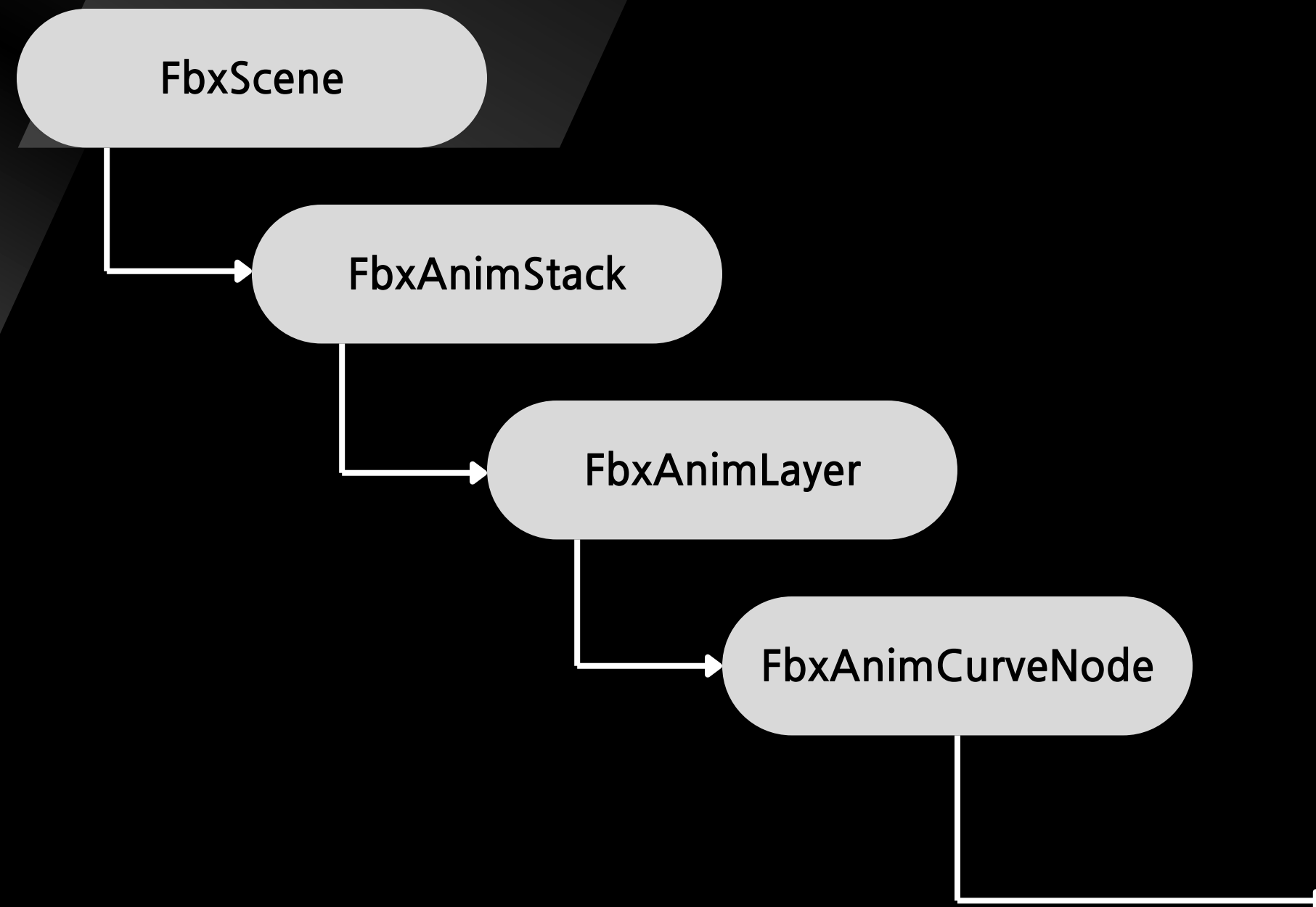
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Animation Sequence



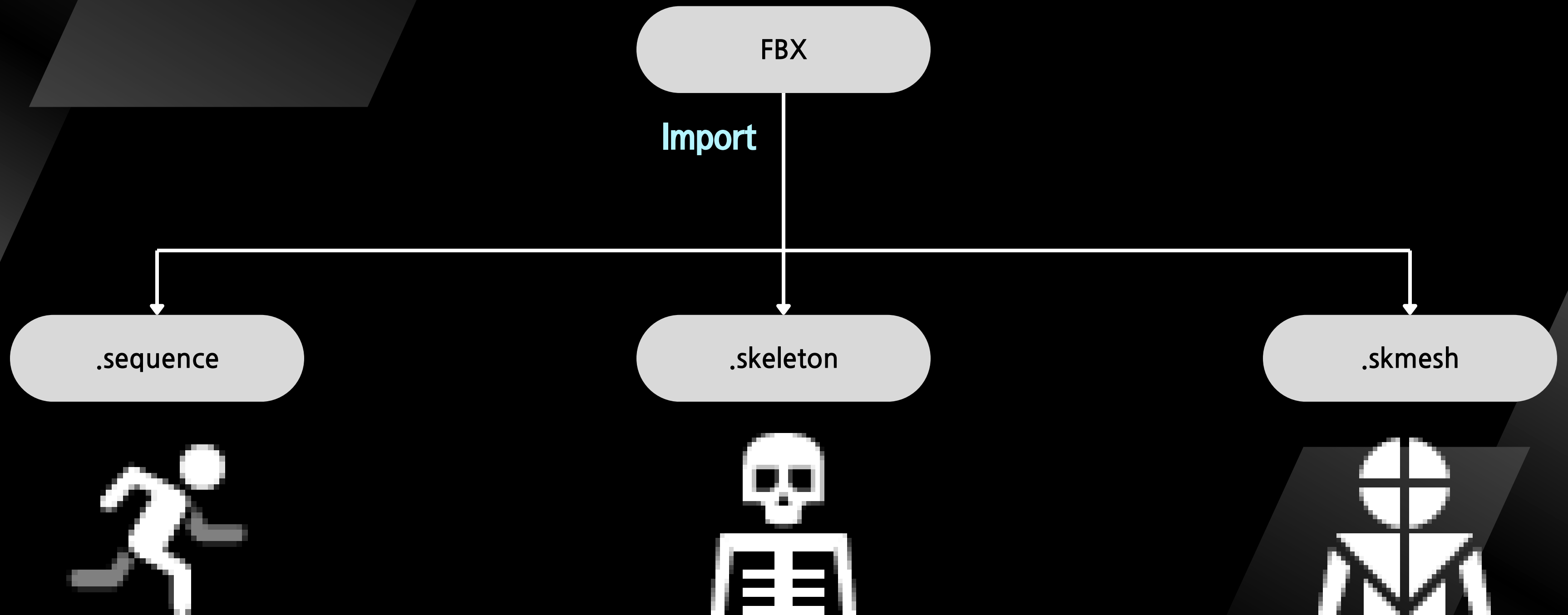
```
class UAnimSequence : public UAnimSequenceBase
{
public:

    UAnimDataModel* DataModel = nullptr;
    FString SkeletonAssetPath;
    USkeleton* Skeleton = nullptr;
};
```

```
class UAnimDataModel : public UObject
{
public:
    TArray<FBoneAnimationTrack> BoneAnimationTracks;
    TArray<FAnimNotifyTrack> NotifyTracks;

    FFrameRate FrameRate;
    int32 NumberOfFrames = 0;
    int32 NumberOfKeys = 0;
    FAnimationCurveData CurveData;
};
```

FBX Asset 분리 구조

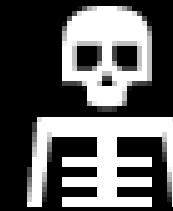


FBX Asset 분리 구조

.manifest

```
{  
  "AnimationSequencePaths": [ "Asset/Animation/MM_Fall_Loop/Unreal Take.sequence" ],  
  "PhysicsAssetPath": "",  
  "SkeletalMeshAssetPath": "Asset/SkeletalMesh/Bin/MM_Fall_Loop.skmesh",  
  "SkeletonAssetPath": "Asset/Skeleton/Bin/MM_Fall_Loop.skeleton",  
  "SourceFileWriteTime": "13423679341",  
  "SourcePath": "Asset/SkeletalMesh/UE_MF/MM_Fall_Loop.fbx",  
  "Type": "SkeletalImportManifest",  
  "Version": 1  
}
```

Skeleton



SkeletonAssetPath

Animation



SkeletonAssetPath

Skeletal Mesh



Animation State Machine

“어떤 Animation Sequence를 재생할지 결정”

Lua Script

```
local sm = self.StateMachine

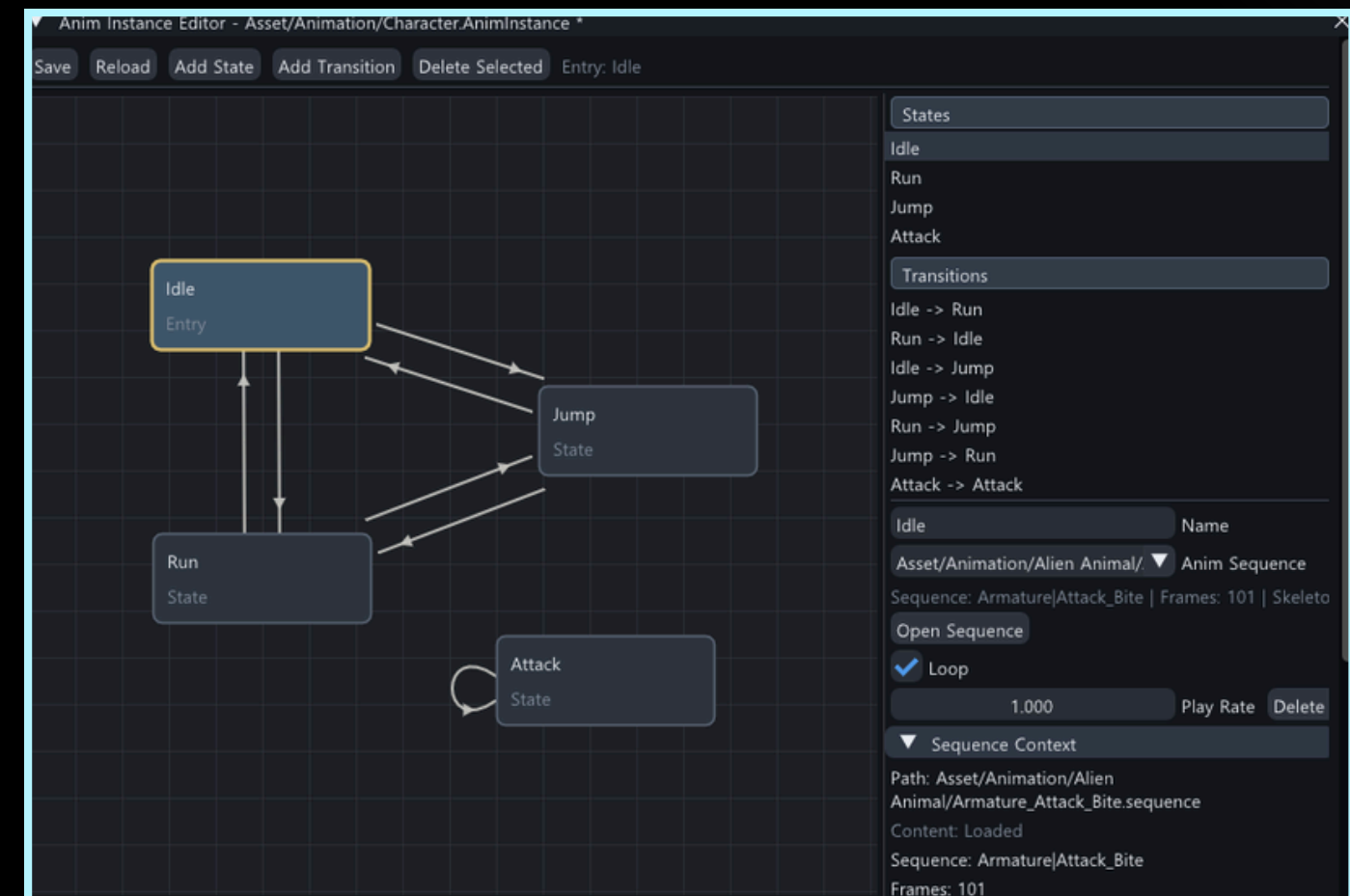
sm:RegisterParameterInt("Direction", Direction.Idle)

addState(sm, "Idle", "Asset/Animation/Alien Animal/Armature_Attack_Bite.sequence", 0)
addState(sm, "Forward", "Asset/Animation/Alien Animal/Armature_Attack_Hit.sequence", 1)
addState(sm, "Left", "Asset/Animation/Alien Animal/Armature_Die_1.sequence", 2)
addState(sm, "Backward", "Asset/Animation/Alien Animal/Armature_Jump.sequence", 3)
addState(sm, "Right", "Asset/Animation/Alien Animal/Armature_Rest.sequence", 4)

sm:SetEntryState("Idle")

sm:AddIntEqualsTransition("Any", "Idle", "Direction", Direction.Idle, 8.0, 0)
sm:AddIntEqualsTransition("Any", "Forward", "Direction", Direction.Forward, 8.0, 0)
sm:AddIntEqualsTransition("Any", "Left", "Direction", Direction.Left, 8.0, 0)
sm:AddIntEqualsTransition("Any", "Backward", "Direction", Direction.Backward, 8.0, 0)
sm:AddIntEqualsTransition("Any", "Right", "Direction", Direction.Right, 8.0, 0)
```

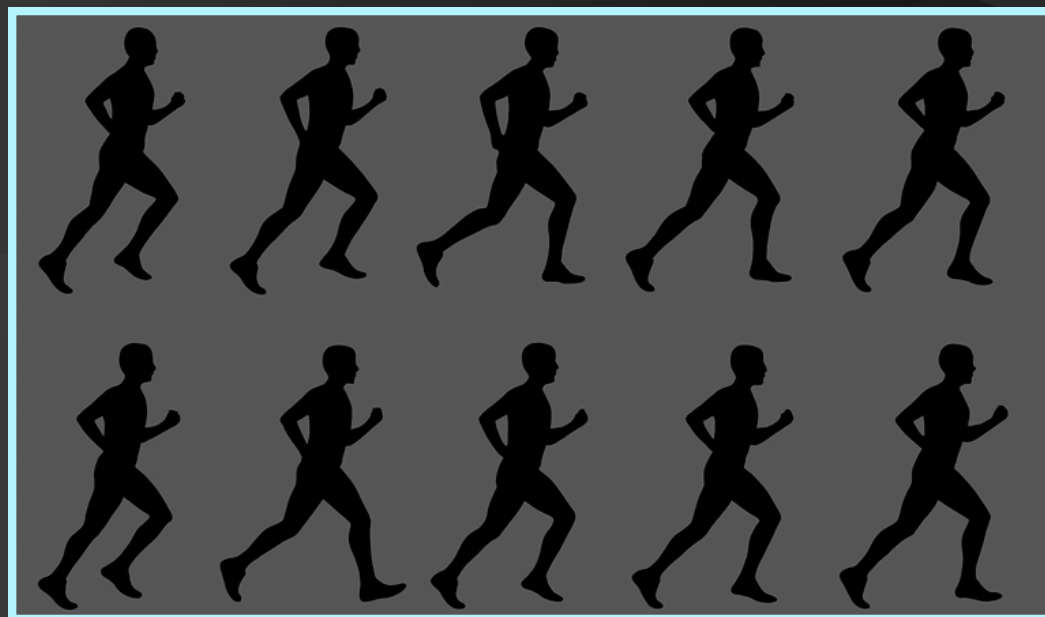
Animation Instance Asset



Animation Instance

“Animation Sequence 재생, 블렌딩”

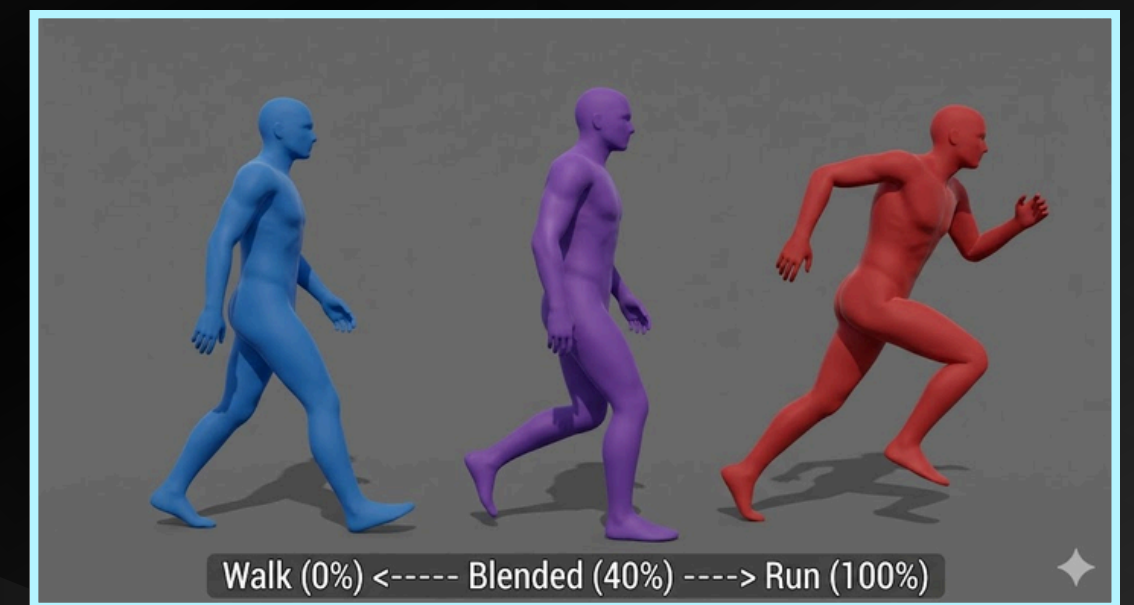
결정된 Sequence



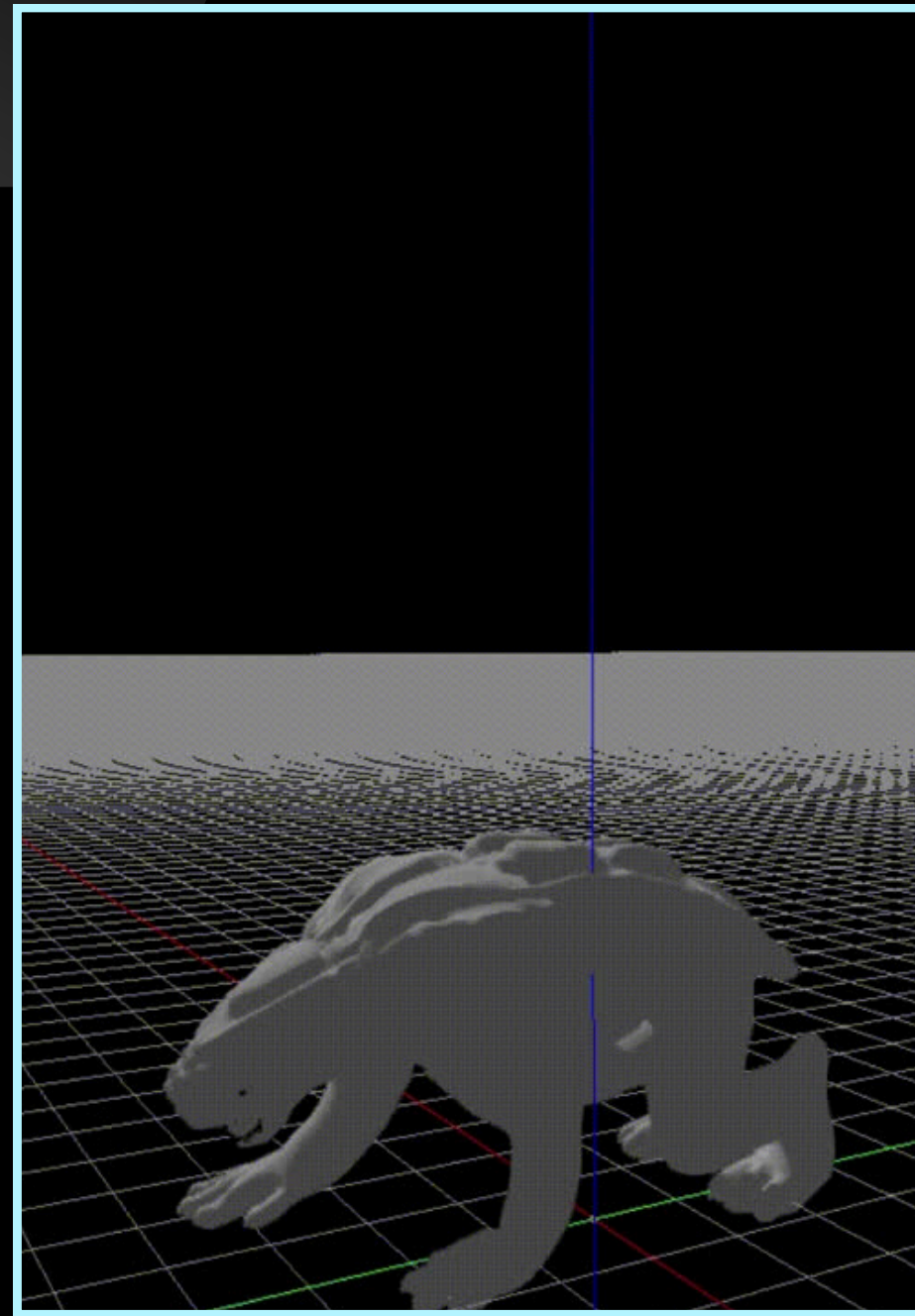
시간 t에서의 Pose 계산



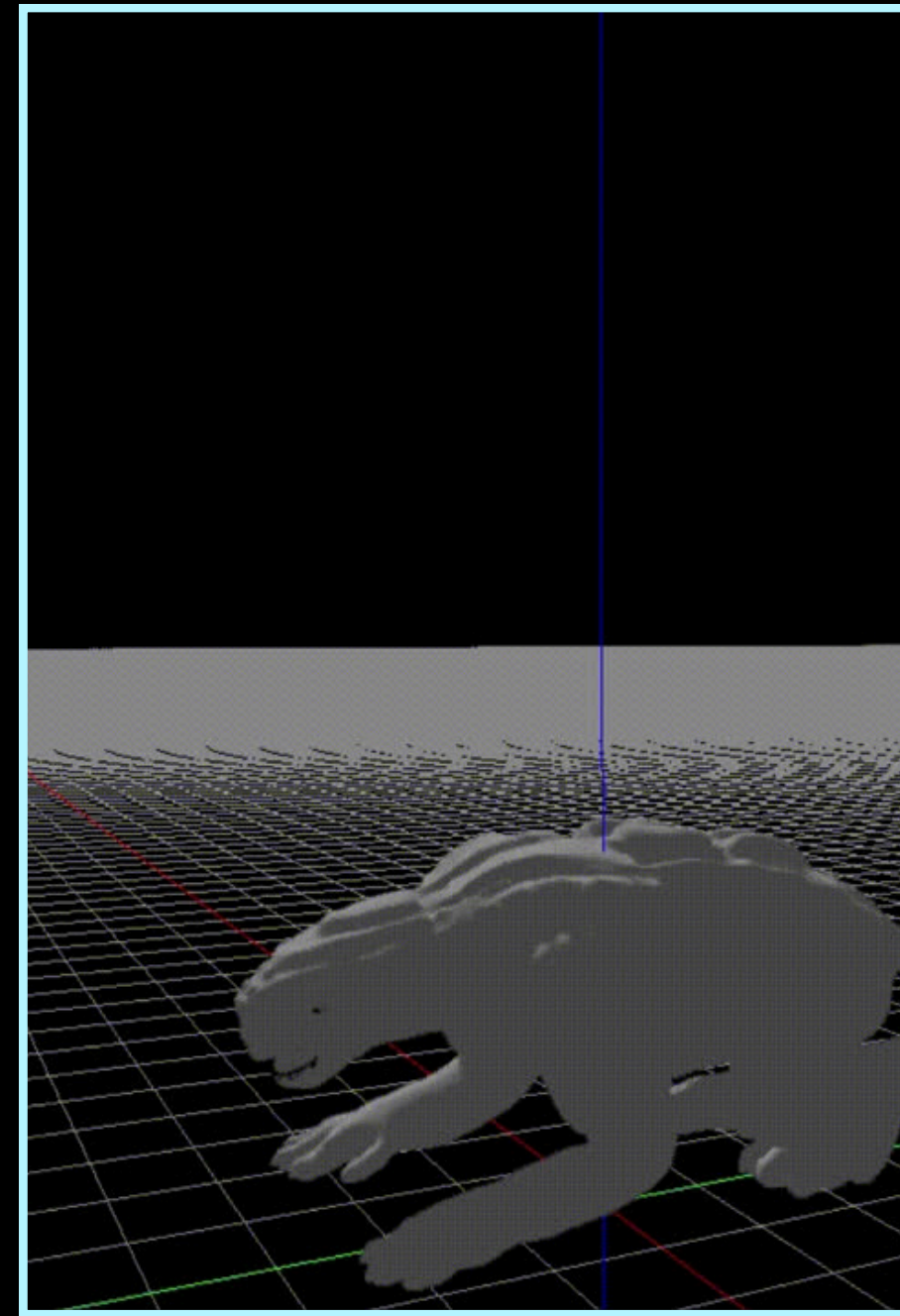
2개의 Pose Blending



Animation Blending

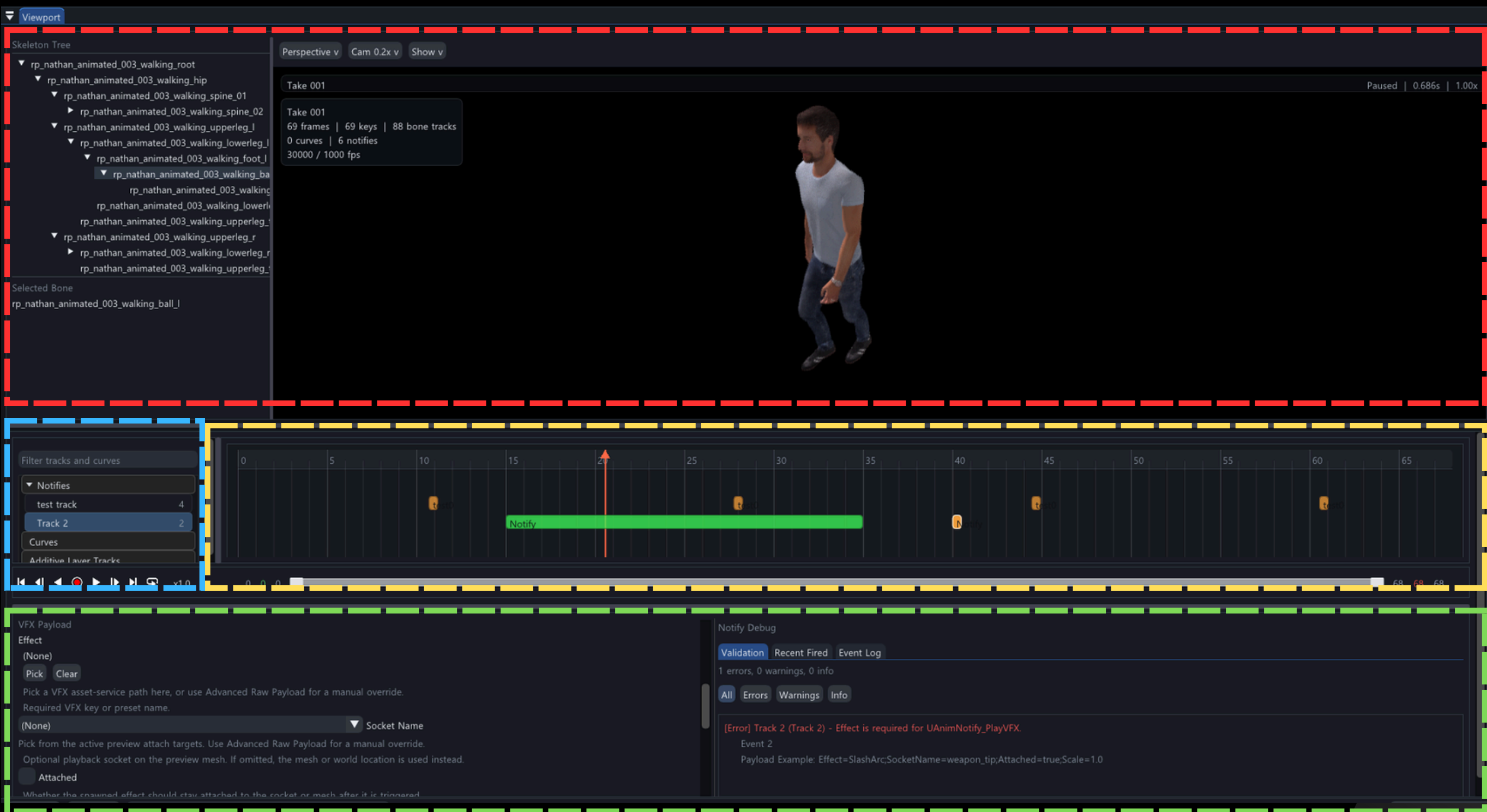


Blend Speed: 3 (약 0.333s)
부드러운 Transition



Blend Speed: 100 (약 0.01s)
Transition 시 끊기는 느낌

Animation Sequence Viewer



Animation Notify



Notify

- 특정 시점에 1회 실행

예: Footstep, Shoot

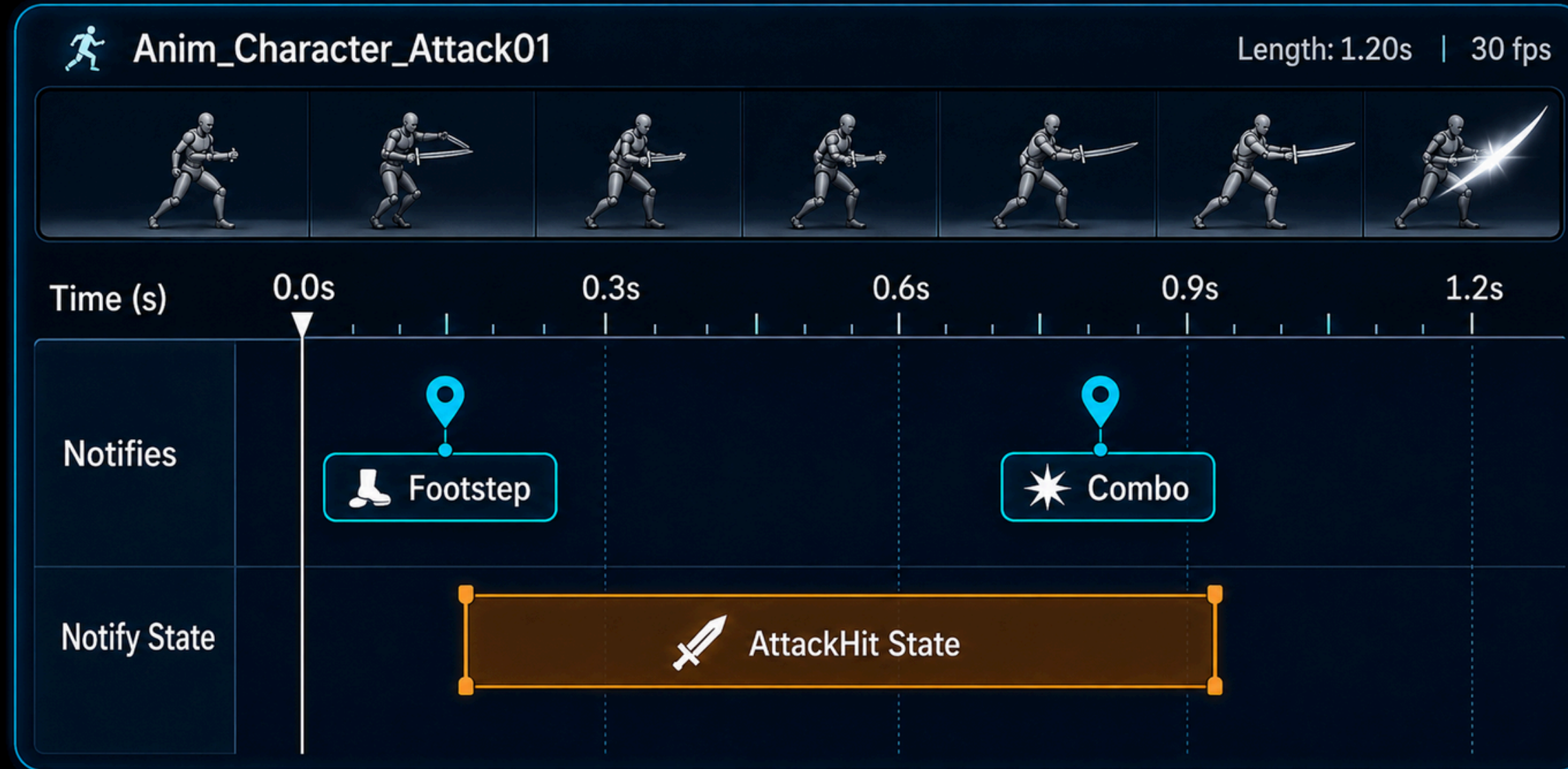


Notify State

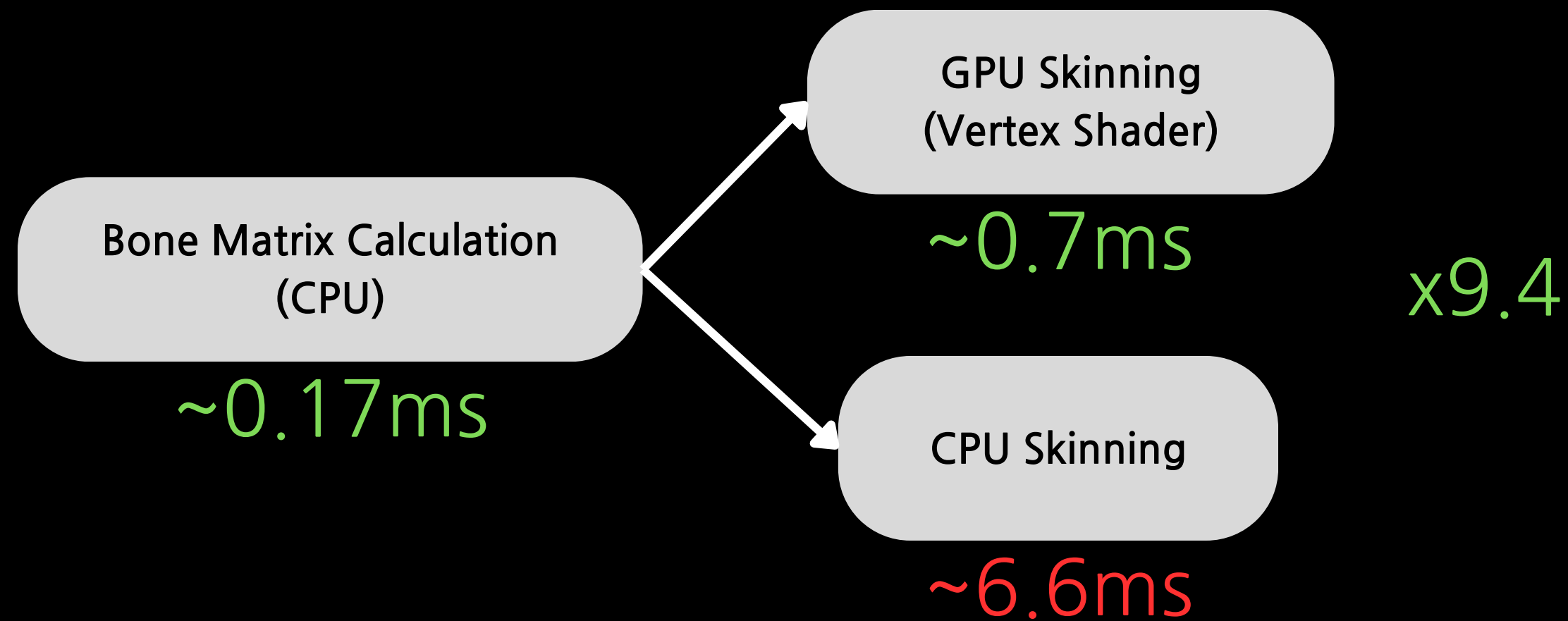
- 시작과 끝이 있는 구간 이벤트

예: AttackHit, ComboWindow

Animation Notify



GPU Skinning



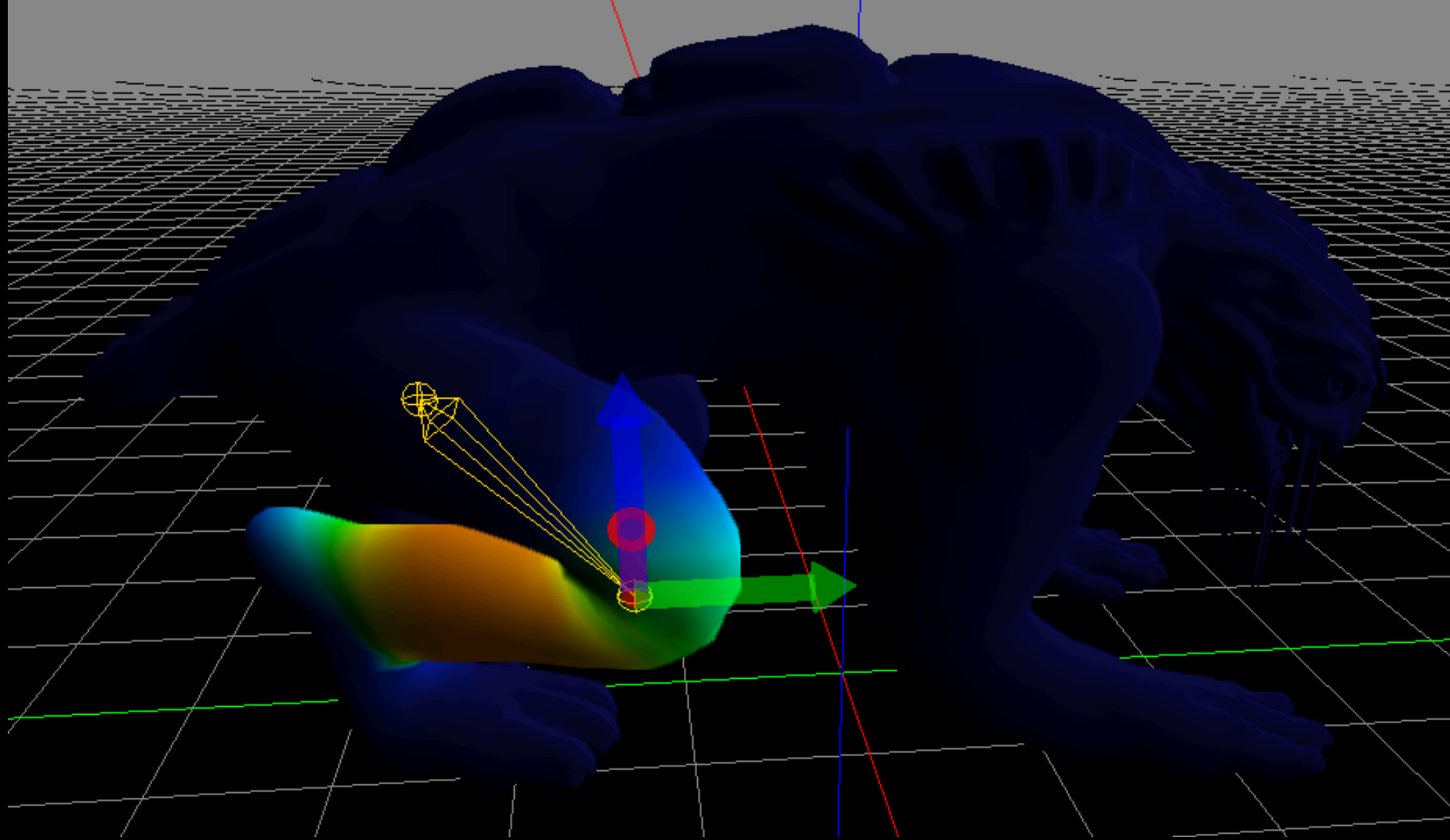
평가 환경

GPU: NVIDIA GeForce RTX 5060

CPU: Intel i7-14700

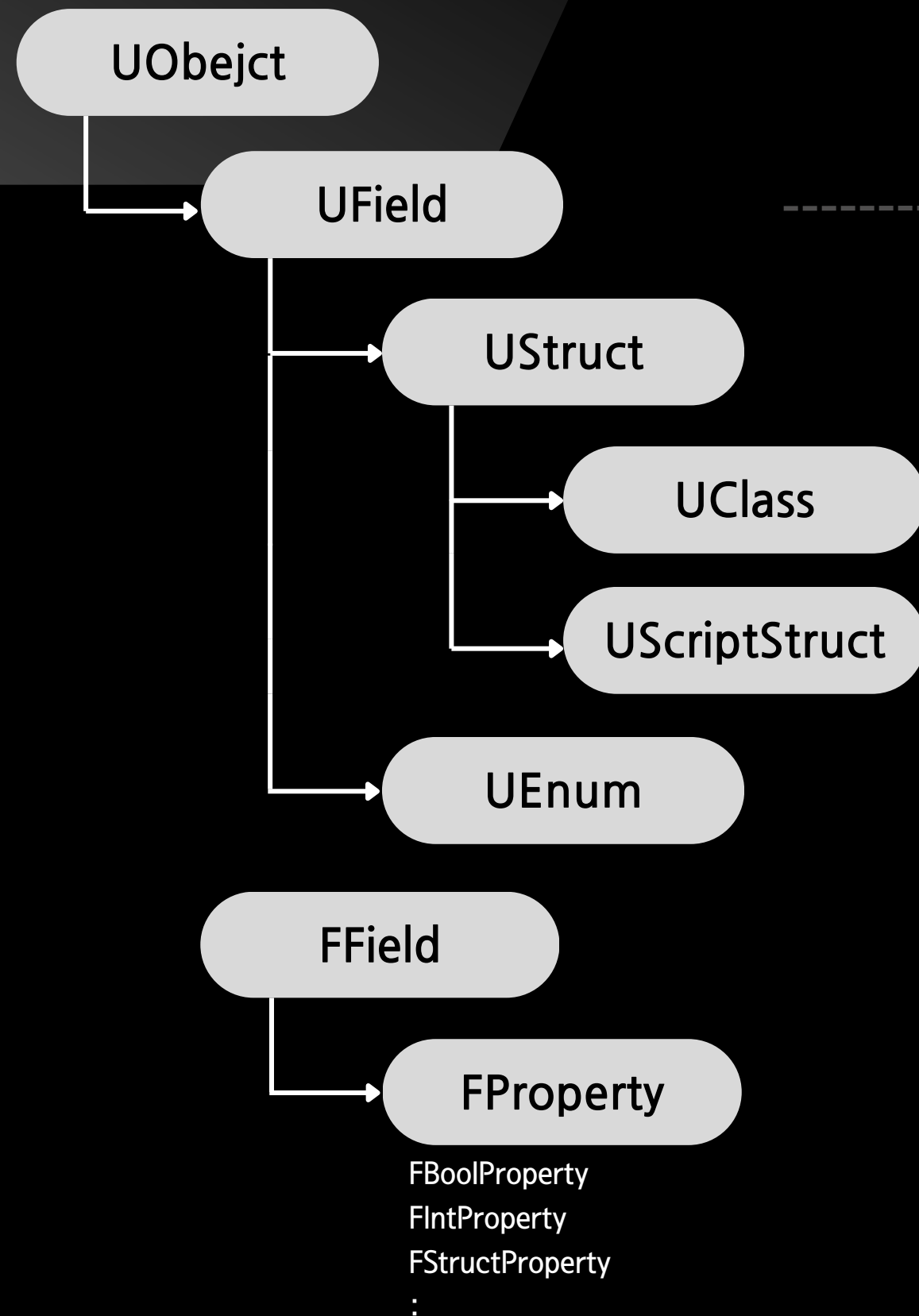
Mesh: 21,506 tri / 12,219 vert

Bone Weight Heatmap



각 Vertex 가 선택된 Bone 의 영향을 얼마나 받는지 시각화

Unreal Reflection



최상위 계층

프로퍼티 배열, 순회 등 공통 기능

C++ Class

C++ Struct

C++ Enum Class

멤버 변수 하나의 메타 타입
Offset, Type, Display Name 등

Unreal Reflection

```
#include "Generated/Object.generated.h"
```

```
UCLASS()
```

```
class UObject
```

```
{
```

```
public:
```

```
    GENERATED_BODY()
```

```
    UObject();
```

```
    virtual ~UObject();
```

```
UPROPERTY(VisibleAnywhere, Transient)
```

```
float CurrentTime = 0.0f;
```

```
UPROPERTY(VisibleAnywhere, Transient)
```

```
float NextTime = 0.0f;
```

```
UPROPERTY(EditAnywhere, Min = 0.0, Max = 1.0, Speed = 0.01)
```

```
float PlayRate = 1.0f;
```

```
UPROPERTY(EditAnywhere)
```

```
bool bLoop = true;
```

Generate.py가 .h 헤더 파일 스캔



UCLASS(), USTRUCT(), UENUM() UPROPERTY() 파싱



*.generated.h 파일 생성



하나의 cpp 파일에 메타 데이터 코드 생성

시연

JISOO ENGINE